

Leveraging Binary Ear Canal Scanning Code for User Authentication in Earbuds

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Abstract

Ear canal scanning and sensing (ECS) have emerged as a novel biometric authentication method for earbuds paired with personal devices. Existing ECS systems rely on “black-box” machine-learning classifiers, facing challenges such as extended enrollment time, large storage requirements, and data leakage risks. To address these issues, we propose EarID, a learning-free ECS biometric system that enhances efficiency while maintaining accuracy. EarID uses masked cepstrum features to capture user-specific in-ear multipath, Bio-Hashing to generate binary ECS codes for reduced communication overhead, and a fuzzy commitment scheme for secure template matching. Evaluations show that EarID achieves 98.1% balanced accuracy, reduces enrollment time by 3.8 $\times$, server storage by 22 $\times$, and communication data by 31.3 $\times$ compared to baseline systems. These results establish EarID as a scalable, secure, and efficient solution for real-world deployment in personal mobile devices.

CCS Concepts

• **Human-centered computing** $\rightarrow$ **Ubiquitous and mobile computing**.

Keywords

Biometrics, Acoustic Sensing, User Authentication

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1 Introduction

Biometric authentication uses human physiological or behavioral patterns to recognize the user, deemed a secure, reliable, and user-friendly authentication system. Based on those biometric modalities, a user only needs to enroll his biometric data and authenticate by presenting the new biometric data samples. Therefore, due to its outperforming security and convenience, biometric authentication has been used in many computer systems, for example, fingerprint authentication on smartphones (TouchID [10]), voice matching for smart home control (Alexa Voice Control [2]), and purchasing by taking a face image (Pay By Face [19]), enabled by fingerprint, speaker, and face recognition algorithms [3, 15, 26], respectively.

Although these biometric solutions have been widely accepted and deployed, their performance requires re-evaluation on advanced platforms [13, 21, 27]. Among them, in-ear wearable devices (e.g., earbuds) are prevalent among customers and are continuously worn by users, making them promising tools for collecting massive and diverse human physiological data to develop biometric systems. Consequently, a new biometric modality that utilizes ear canal scanning/sensing (ECS) has come into the spotlight. With the earbud’s embedded speaker and inward-facing microphone placed inside the occluded ear canal (see Fig. 1), the ear canal and middle ear structures are scanned in a manner similar to radar and sonar systems. Early clinical and acoustic research [16, 22] has identified that different ECS responses reflect user-level uniqueness. Therefore, capturing these unique ECS responses is key to building an authentication system.

Recently, many user authentication systems have used the ECS and its FFT feature as a new biometric solution just using earbuds [5, 7, 9, 14, 23, 25]. For instance, Mahto *et al* in [14] studies the advantages of using human inaudible frequency ranges during the ear canal scanning. EarDynamic [23] operates at the inaudible frequencies, and exploits the ear canal deformation as extra discriminative

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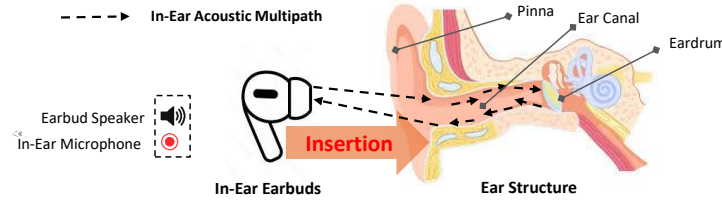


Figure 1: Illustration of ear canal scanning/sensing.

features. HeadFi [7] extends the in-ear sensing capacity to low-end devices without an in-ear microphone. Notably, the performance benchmark can be found as up to 97.53% balanced accuracy (BAC) reported in [9] and as 0.3% low equal error rate (EER) in [25]. While existing works focus on different aspects of ear-scanning technologies, limited studies are targeting authentication decision-making algorithms. Although machine learning classifiers, adopted by most works, show the ability to learn from the FFT feature with high accuracy, the challenges of complex training, large model size, and privacy concerns of sharing biometric data hinder its practical usage. In addition, training in a “black-box” manner might also lead to interpretability problems.

To address these issues, this paper differs from existing works by proposing an ECS biometric remote authentication system that does not rely on training machine learning classifiers as the authentication algorithm. We introduce EarID, a novel ear-scanning biometric system utilizing a fuzzy commitment scheme [12]. EarID leverages cryptographic technologies to store user enrollment data as binary keys and commitments, addressing data storage overhead and privacy concerns. As another contribution, we propose a compact, user-specific cepstrum feature representation with biometric-informative masking, emphasizing the most unique feature subset that can be recognized for a specific user. This feature can be further transformed into a fixed-length binary ear scan code (or ECS code) shared between the client and server and integrated into the fuzzy commitment template. Additionally, a cryptographic nonce can be included in the binary ear scan code to prevent replay attacks. We also developed ECS data collection hardware and software platforms for ear-scanning data collection and processing. Evaluated on the collected data, EarID demonstrates comparable accuracy to its classifier-based counterparts but achieves greater efficiency in enrollment time and data storage.

2 Background

Ear Canal Scanning Front-End. Instead of directly measuring the middle ear impedance, many user authentication papers [5, 7, 9, 23] scan the ear canal similar to radar and sonar systems. The ear canal scanning front-end hardware requires a transmitter (earbud speaker) to emit an acoustic waveform into the ear canal and simultaneously record the response with a receiver (in-ear microphone), as shown in Fig. 1. When the earbuds are inserted firmly into the ear canal displaying a wide-band waveform, the reflected stimulus waveform is then captured by the in-ear microphone. The characteristics of the acoustic channel can be expressed using cross-correlation with the waveform, expressed as

$$h(n) = \sum_{t=0}^{\infty} s(t)r(n-t), \quad (1)$$

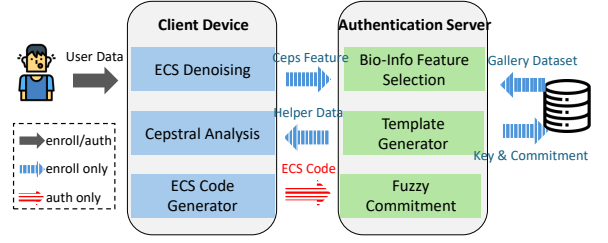


Figure 2: The system overview of proposed EarID.

where $h(n)$ is the target in-ear impulse response at sample index n , $s(t)$ is the transmitting waveform at time t , and $r(t)$ is the reflected signal captured by the in-ear microphone. Nevertheless, the magnitude transfer function (FFT features), which is the magnitude of the Fourier Transform of $h(n)$, is employed by most papers due to the smaller intra-user variation by removing the noisy phase information. The existing works focus on improving the ear canal scanning front-end, including the choice of waveform, the impact of re-insertion, the effect of device frequency responses, and the impact of ear canal deformation.

Decision Making Back-End. Using machine-learning classifiers is mostly found in the current ECS biometric authentication studies, including Support Vector Machine (SVM) [7, 9, 25], Linear Discriminant Analysis (LDA) [14], and ensemble learning methods [23]. The classifier can be trained for either multi-class classification, where each user is assigned a unique class label (known as user identification), or binary classification, where each user trains a binary classifier to distinguish between the legitimate user (positive label) and random users (negative label), known as user verification¹. Meanwhile, template matching is another authentication back-end commonly used for fingerprint and iris authentications. Template matching differs from classifiers, by shifting the task of learning unique features from the classifier to the feature extractor, and it only requires an efficient matcher to verify whether the current user is the legitimate user. It also benefits from existing template protection technologies designed for biometric security and privacy. However, no existing paper has developed an approach using template matching. The major challenge is that template matching requires a fixed-length (compact) feature representation, which must be informative enough to recognize the specific user and invariant to multiple data captures from the same user. In this paper, we tackle this challenge and propose EarID, a complete system design for a template-matching-based ECS biometric system with a fuzzy commitment scheme, using new biometric-informative representations.

3 EarID System Overview

3.1 Framework

Illustrated in Fig. 2, the EarID system consists of two main components: the client device and the authentication server, connected via a secure communication channel. The client device is equipped with earbud speaker(s) and in-ear microphone(s) capable of performing ear canal scanning. Additionally, it contains a computational unit for signal and feature processing, which runs on the earbuds or

¹In this paper, we focus solely on user verification when mentioning authentication, as user verification is more common than identification in head-wearable devices.

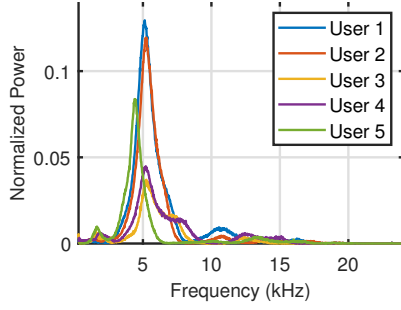


Figure 3: The ECS frequency responses from 5 users cover the normal audible frequencies using commercial devices.

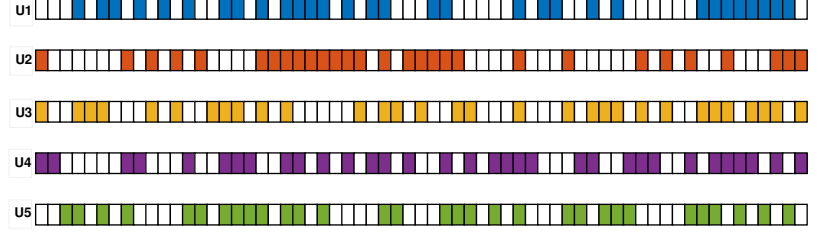


Figure 4: The binary ECS codes transformed from frequency responses from the same 5 users, preserving the uniqueness among users.

their paired device, such as a smartphone. The authentication server, with greater computational power, maintains the gallery dataset from the population, enrolls new users, and stores their keys and commitments in the enrollment database.

Client Device. The role of the client device is to perform ear canal scanning and transform the raw data into a compact and informative representation for EarID authentication, i.e., a fixed-length binary ECS code. Aided by the helper data from the authentication server during the enrollment stage, a user-specific ECS code can be generated locally when the legitimate user is wearing the earbuds. While the binary ECS code is unique and compact—making it suitable for wireless transmission—the raw ear scanning data never leaves the user’s device, protecting the raw biometric data from potential leakage.

Authentication Server. The authentication server is responsible for the following: (1) During enrollment, maintaining an ever-developing gallery dataset, the server evaluates how “unique” the new user is using a biometric information metric; (2) it generates the user template and helper data—the former is stored for matching, and the latter is sent to the client device to reproduce the ECS code matched with the template; (3) During authentication, template matching is performed based on a fuzzy commitment scheme to avoid using raw biometric data.

3.2 Feature Transformation

There are three crucial feature transformation steps in our proposed EarID: (1) converting the impulse response in Eq. (1) (or equivalently, its FFT) into cepstral features, which are less correlated and emphasize physical multipath information; (2) selecting a “unique” cepstral feature subset specific to a user by computing biometric information using the user’s data and the gallery data; (3) binarizing the selected subsets to generate the ECS code (for authentication) and the fuzzy commitment template (for enrollment). The goal of these transformations is to reveal and compress the uniqueness of the raw ECS estimations shown in Fig. 3. The results of the feature transformation in EarID are shown in Fig. 4, where each user’s ECS code is presented in binary format. This format allows for efficient and fast data storage and transmission and directly reflects differences between users.

Cepstral Analysis. The Fast Fourier Transfer (FFT) of estimate in-ear impulse response in Eq. (1) can be highly correlated and redundant. We aim to compress this ECS representation with physical multipath information. Therefore, we apply cepstral analysis to the FFT feature vector, which reveals multipath information such as echoes and reverberations in the quefrequency domain [17]. This process generates a new cepstrum feature vector whose entries are uncorrelated and can be interpreted as unique multipath reflections within the ear canal. The cepstrum feature is defined as:

$$c(n) = \mathcal{F}^{-1} \{ \log |H(f)| \}, \quad (2)$$

where $c(n)$ denotes the cepstrum and $\mathcal{F}^{-1}$ is the inverse Fourier Transform. Because the FFT feature is a real-value vector, we use the inverse Discrete Cosine Transform (IDCT) instead. Each entry of the transformed feature $c(n)$ represents a unique multipath strength from the ear canal, where n indicates the time of flight. Given the ear canal length and the speed of sound, we can safely remove cepstral components outside the reasonable time-of-flight range (e.g., 10 ms). This step known as time-of-flight liftering [17] reduces the feature dimension while removing irrelevant noises.

Biometric-Informative Feature Masking. The authentication server must identify the informative multipath from a user’s enrollment cepstrum features and discard the less informative ones to compress the data. This “information” is considered high if a feature exhibits low intra-subject variation and high inter-subject variation. Using handwriting recognition as an analogy, Alice’s distinct handwriting style makes her writing easily recognizable, and her handwriting remains consistent across different instances. To this end, we utilize the *biometric information* (BI) defined by Adler et al. [1] as a metric, which is the distribution divergence between the feature distribution from the user enrollment data and that from the population. We modify the original definition using the Rényi divergence as:

$$D_\alpha(P||Q) = \frac{1}{\alpha - 1} \log_2 \left(\sum_{x \in \mathcal{X}} P(x)^\alpha Q(x)^{1-\alpha} \right), \quad (3)$$

where $P(x)$ represents the probability distribution of the user’s feature, $Q(x)$ represents the probability distribution of the population’s feature, and α is the order of the Rényi divergence. This modification allows us to focus on different aspects of distribution divergence by tuning α . The empirical study shows that when

Algorithm 1: Pseudocode for RRFC

Input: Enroll ear scan code $\mathbf{b} \in \mathbb{B}^{n \times 1}$, fresh ear scan code $\mathbf{b}' \in \mathbb{B}^{n \times 1}$, cryptographic nonce $\mathbf{n} \in \mathbb{B}^{m \times 1}$, ECC_Encode, ECC_Decode

Output: Commitment $\mathbf{c} \in \mathbb{B}^{n \times 1}$, key $\mathbf{k} \in \mathbb{B}^{k \times 1}$, authentication decision $D \in \{0, 1\}$

- 1 **Function** Enroll($\mathbf{b}$):
- 2 (1) Generate random key $\mathbf{k}$; (2) Generate codeword: $\text{code} = \text{ECC_Encode}(\mathbf{k})$; (3) Create commitment: $\mathbf{c} = \mathbf{b} \oplus \text{code}$; (4) Store $\mathbf{c}$ and $\mathbf{k}$ on the server;
- Result:** $\mathbf{c}, \mathbf{k}$
- 3 **Function** Authenticate($\mathbf{b}', \mathbf{n}$):
- 4 (1) Generate and transmit nonce $\mathbf{n}$; (2) Receive fresh ear scan code $\mathbf{b}'$; (3) Retrieve stored commitment $\mathbf{c}$;
- 5 (4) Combine: $\text{code}' = \mathbf{c} \oplus \mathbf{b}' \oplus \mathbf{n}$; (5) Decode key: $\mathbf{k}' = \text{ECC_Decode}(\text{code}')$; (6) $D = 1$ if $\mathbf{k}' = \mathbf{k}$ else 0;
- Result:** D

$\alpha \rightarrow 0$, the EarID system achieves the best overall performance. Next, it is straightforward to generate a feature mask, select cepstrums with relatively high BI, and discard those with low BI that may lead to errors. However, setting a hard BI threshold for all users is not desirable. For instance, if a user has uniformly high BI cepstrums, only a subset of the highest BI features is sufficient. Conversely, for a user with generally low BI, the system should still attempt to recognize them using their most informative cepstrums. Therefore, we propose the masking approach based on each user's BI across cepstrum features by automatic thresholding using Otsu's method [18]. The resultant user-specific binary feature mask is then applied to the cepstrum feature, where '1' means the high-BI cepstrum to retain and '0' means the low-BI cepstrum to discard.

BioHashing into Binary ECS Code and Template. Finally, the selected cepstral feature subset with high biometric information is transformed into a fixed-length binary representation for efficient computation, storage, and communication. Specifically, this step involves BioHashing [11], a random matrix method that projects the masked feature subset onto a higher dimension (e.g., 255) and uses the sign to represent each vector entry (positive as '1', negative as '0'). The derived binary code (with the random matrix as helper data) preserves uniqueness, exhibiting high intra-subject similarity (low bit-error rate for the legitimate user) and high inter-subject variation (high bit-error rate for intruders). Together with the feature mask from the previous step, this random matrix (fixed after generation) serves as the helper data that must be sent from the server to the client to reproduce the ECS code for the legitimate user.

3.3 Enrollment and Authentication

Now, with the binary code representing the ECS features, we can utilize the Fuzzy Commitment Scheme [12] to eliminate the need for training a machine learning classifier. Furthermore, to prevent code interception and replay attacks, we incorporate a time-sensitive nonce into the commitment (ECS code). This step, called Replay-Resistant Fuzzy Commitment (RRFC), is detailed in Algorithm 1. The "ECC_Encode" and "ECC_Decode" operations in the algorithm

Table 1: Audio and EarID Design/Evaluation Parameters

Acoustic Parameter	Value	Acoustic Parameter	Value
Volume	user adjust	Total Number of User	44
Sampling Frequency	48 kHz	Scan per Session (Enroll/Auth)	20/5
MLS Duration	1 second	Sessions to Enroll/Auth.	2/1
Chirp Frequency Range	1 kHz - 8 kHz	Total Enroll/Test Sessions	2/4
Chirp Duration	1 second	Quiet Room SPL	30-40 dBA
FFT/Ceps Number	1024/256	Indoor/Outdoor Noise	~60/70 dBA ²

refer to the encoding and decoding processes of an error-correction code, such as Bose–Chaudhuri–Hocquenghem (BCH) codes [4], which are employed in the rest of this paper. The resulting codeword is combined with the ECS template using a bit-wise XOR operation to create a commitment. In this way, the user enrollment data (i.e., the commitment and key) can be securely stored at the authentication server. Even if the enrollment database is compromised, the key and commitment reveal no information related to the raw biometric sample, thereby protecting the user's privacy and security. Moreover, the use of an error-correction code enables the system to tolerate small random errors that may occur during legitimate attempts.

4 Data Collection

Hardware and Software. Although many high-end ear-wearable devices, such as noise-canceling earbuds, are equipped with in-ear microphones, there are no development tools available to access those microphones directly. Therefore, we manually insert a small-sized microphone (CMC-3015-44L100 [6]) inside each foam tip of commercial off-the-shelf (COTS) earbuds (Sony MDREX15AP In-Ear Earbud [20]). The microphone is positioned inside and faces towards the ear canal when worn by the user. To simultaneously capture responses from both ears, all audio outputs (earbud speakers) and inputs (in-ear microphones) are connected to a full-duplex USB audio codec. The hardware platform is shown in Fig. 5. Additionally, we developed the EarID software using Matlab App Designer for data collection and authentication evaluation, as shown in Fig. 6 and Fig. 7.

Subject Recruitment and Ethic Consideration. This experiment involves human subjects and is approved by IRB protocols. All subjects were recruited on the university campus, ensuring sufficient diversity in terms of gender, race, age, and including students, faculties, and staffs. In total, we recruited 44 users to create our user pool. Compared to existing studies, the size of our user pool exceeds the median size (26) of previous works [5, 7, 9, 14, 23, 25], and is second only to the largest user pool size (52) in [25]. Since the subjects are only required to wear the earbuds and listen, the data collection poses no harm to them. To mitigate risks associated with user privacy, all participants were fully informed about the data collection process and provided their consent before participating in the experiment. Although the biometric features need to be annotated for the authentication system design, we use the non-sensitive "last four digits of their phone number" to replace sensitive identifiers (e.g., name, ID). Additionally, privacy information unrelated to this study, such as health conditions, was not recorded during the experiment.

²This experiment is for future evaluation and discussed in Section 6

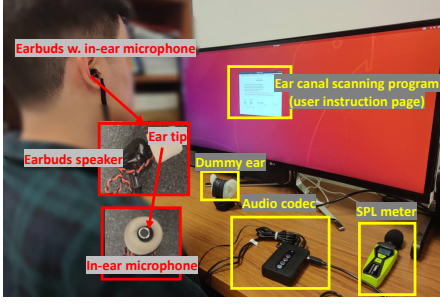


Figure 5: EarID Testbed Data Collection.

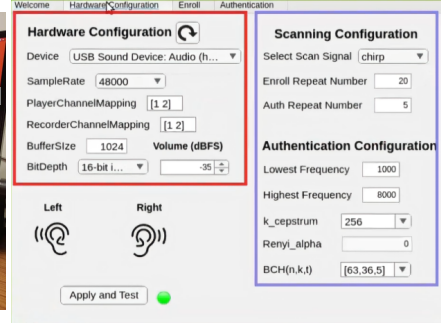


Figure 6: EarID Algorithm Parameters.

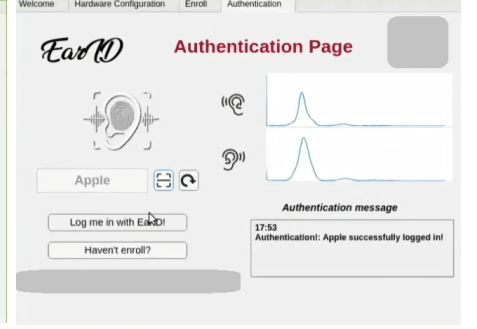


Figure 7: User Authentication Page.

Ear Scanning and Recording. Human subject tests were conducted in a quiet room (30–40 dBA). In each session, users were instructed to wear or re-wear the earbuds until they were fully occluded, remain silent, and behave naturally. With the parameters set as shown in Table 1, users adjusted the volume to a suitable level and clicked the “scan” button. The system then transmitted chirp signals and maximum-length sequences (MLS) as excitation signals, each repeated 20 times during enrollment sessions. These two excitation methods yielded nearly identical results. To enhance system performance, it may be possible to reduce the number of repetitions while increasing the number of enrollment sessions; we leave this for future work. For authentication, since the user had already worn the earbuds in the current session, only data from that session was required, with five repeated scans. In total, the data recording process included two enrollment sessions and four test sessions. Additionally, we recorded data under indoor and outdoor noise conditions at approximately 60 dBA and 70 dBA, respectively, for future evaluation, as discussed in Section 6. In this paper, we do not present the results of those participants tested in different conditions, such as cross-day evaluations.

5 Evaluation

To evaluate the performance of EarID, we focus on authentication accuracy and system overhead. For the evaluation, we conducted all experiments over 20 trials and reported the aggregated results. In each trial, we randomly split the entire dataset into a gallery dataset and a new user’s data. For users in the gallery dataset, all multi-session data were included (enrollment+authentication data). For the remaining new user’s data, enrollment and authentication samples were further selected from the corresponding data collection sessions. We rigorously ensured that the enrollment data and authentication attempts were split in chronological order based on the data collection process. In other words, the enrollment data did not include any future collected samples.

5.1 Accuracy

Bit-wise Error vs. Error-Correction Code. The key to EarID’s accuracy is the XOR operation combining the stored commitment c with the newly generated ear scan code b' in Algorithm 1. Specifically, the bit error is the bit-wise difference between the ECS code and template, defined as $e = b \oplus b'$. It is worth noting that we do not consider the cryptographic nonce in this evaluation. Equivalently,

False Rejection Rate (FRR) becomes the probability that the total number of bit errors exceeds the error correction capacity ($\sum e_i > t$) given legitimate attempts, while False Accept Rate (FAR) becomes the probability that the total number of bit errors is less than the error correction capacity ($\sum e_i < t$) given unauthorized attempts.

Results. The accuracy performance results are shown in Table 2. We tested classifiers, including linear/quadratic SVM, decision tree, and 3-layer Multilayer Perceptron (MLP), and compared them with the proposed EarID using three ECC configurations: BCH(63,36), BCH(127,64), and BCH(255,131) with the capacity to correct 5/10/18 bit errors, respectively. The training and evaluation strategy for classifiers follows that in [9]. Each user trains their own classifier by combining their data (1) with gallery user data (9) and evaluates it with their own presence and intruders (10). Table 2 presents the aggregated results across all users and multiple trials. In contrast, the proposed EarID does not require selecting gallery users in the training data. Instead, it directly compares population feature distributions from all available gallery data using Eq. (3). We observed that all evaluated ECS authentication systems achieve comparable Balanced Accuracy (BAC). Among these designs, all EarID configurations show better BAC compared with the classifiers. EarID-BCH(255,131), especially, outperforms others with the highest BAC of 99.0%. These results demonstrate the good accuracy performance of the proposed EarID system, and a bigger code size leads to better performance.

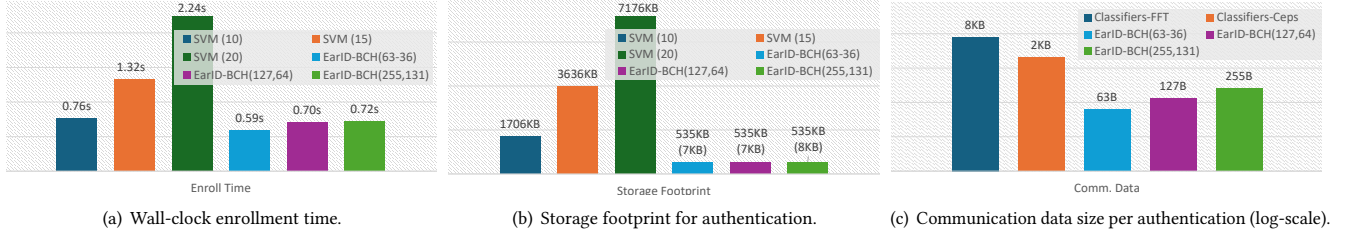
5.2 System Overhead

System overhead in ear canal scanning-based authentication is evaluated by enrollment time, storage footprint, and communication data size between the client device and the authentication server. These factors become particularly significant during the biometric warm-up stage, as the system scales with an increasing number of enrolled users and requires frequent updates. To ensure a fair comparison, we continue to use SVM models for classifier training, scaling up the number of training users to 10, 15, and 20. In contrast, the proposed EarID, which remains insensitive to changes in the number of entries in the gallery dataset, maintains fixed system overheads, as depicted in Fig. 8.

Measurement. The experiments are conducted on a Linux desktop with an Intel(R) Core(TM) i7-9700K CPU @ 3.60GHz and 16 GB RAM. Since the evaluation program uses Matlab, the wall-clock

Table 2: Authentication Performance Comparing Different Classifiers and Template Matching Methods

Metrics	Classifiers				Template Matching		
	Linear SVM	Quadratic SVM	Decision Tree	MLP	EarID-BCH (63,36)	EarID-BCH (127,64)	EarID-BCH (255,131)
FAR (↓)	10.4%	2.5%	0.7%	6.5%	3.1%	3.4%	1.8%
FRR (↓)	0%	7.4%	6.7%	1.9%	0.7%	0.3%	0.2%
BAC (↑)	94.8%	95.0%	96.3%	95.8%	98.1%	98.2%	99.0%

**Figure 8: The system overhead of EarID compared with SVM classifier.**

runtime and the storage footprint are measured using the Matlab Profiler and its built-in functions. It is worth noting that we remove unnecessary data from the classifier models for fair evaluation, such as training data, class labels, optimization state, etc.

Results. We first compare the time users need to enroll, shown in Fig. 8(a). The numbers, e.g., SVM(10), indicate that the SVM classifier is trained using data from 10 users (1 legitimate and 9 gallery users). As the number of users involved in training increases, the enrollment time also increases accordingly. In contrast, EarID systems are evaluated with a fixed set of 20 gallery users, similar to SVM(20), and they learn population information through distribution rather than individual user samples. As a result, the enrollment time for all EarIDs remains nearly constant regardless of the gallery data size. For example, compared to SVM(20), the enrollment time of EarID-BCH(63,36) is 3.8× faster. Regarding storage footprint, EarID only stores helper data on the client side and [key, commitment] pairs on the authentication server, which is significantly smaller compared to machine learning classifier models. In Fig. 8(b), scaling up to 20 users results in 7176 KB storage usage with SVM classifiers, whereas EarID only uses 535 KB (including 7-8 KB for server-side enrollment data), making it 13.4× smaller. Furthermore, conventional classifiers typically train on FFT features and may adapt to cepstrum features. The communication data size required to transmit these features is in kilobytes (KB), whereas EarID systems only require approximately 255 bytes (B) for binary ECS codes, offering a 31.3× efficiency advantage.

6 Discussion

Impact of Biometric-Informative Masking. The purpose of biometric-informative masking is to filter out cepstral features that lack uniqueness or exhibit high intra-subject variability. To study its impact, we excluded this module and generate ECS code for matching. Without biometric-informative masking, the FAR dropped to 0%, but the FRR soared to 86.9%. This demonstrates that the proposed feature selection module effectively identifies useful cepstral features, significantly improving EarID system performance.

Selection of Error Correction Code. We tested only BCH as the error correction code (ECC) for EarID, while other options including Reed-Solomon (RS) [24] and Low-Density Parity-Check (LDPC) [8] codes are more complex to perform. This selection can be optimized in future work, as EarID relies on ECC to tolerate random bit errors caused by variations in the user’s ear canal and the way earbuds are worn. It is noteworthy that the *flexible* (yet sufficient) error-correction capacity, rather than a high error-correction capacity, benefits EarID. Additionally, complexity is an important consideration when implementing ECC on earable devices. Nevertheless, ECC is fundamental for wireless systems and might be utilized without incurring extra costs.

System Update as User/Data Scale Up. As EarID authentication scales up from its warm-up stage (e.g., 20 built-in user data), it continuously collects more data to enhance system accuracy and robustness. With increasing user consent to contribute their data for system development, EarID benefits from more precise population data, improved statistical modeling, and a better understanding of data distribution. Consequently, feature transformation becomes more effective, as the estimation of “biometric information” is more precise, leading to the generation of higher-entropy ECS codes that better capture user uniqueness. Moreover, these improvements impose minimal additional costs on client devices (e.g., earbuds), as the signal processing pipeline remains unchanged and only processes the user’s own data. Only when transitioning to a higher-entropy code, such as EarID-BCH(255,131), does it introduce a negligible overhead, see Fig. 8. In such cases, thanks to EarID’s learning-free and low-complexity design, existing users can simply re-enroll with the help of authentication server’s updated gallery dataset to enhance performance. System updates can ultimately be phased out once the population feature distribution converges.

6.1 Future Work

Robustness and Security Evaluation. In future work, we will conduct a comprehensive evaluation of EarID, focusing on both robustness and security. The robustness of EarID will be assessed

against false-triggering events (e.g., placing earbuds on a table), noisy environments (e.g., indoor/outdoor noise), multiple sessions (e.g., re-wearing earbuds), and multi-day evaluations (e.g., aging effects). **It is important to note that these evaluations involve conditions that are not present in the current user dataset.** Therefore, additional data collection will be necessary to thoroughly test the developed EarID system. For security evaluation, EarID will be tested against impersonation attacks (e.g., legitimate users vs. intruders, silicon ear models, gallery data users), code guessing attacks, and other potential threats.

Optimal Ear Scan Time and Notification Level. We find that the duration of the ear scan and the number of scans per session (see Table 1) in this ongoing work are redundant and can be optimized without significantly degrading authentication performance. Intuitively, reducing enrollment and authentication time can enhance user experience; however, the volume and frequency used for ear scanning may also introduce notification to user. Specifically, if the system operates in an inaudible frequency range, it can be non-intrusive but lacks a notification cue. Conversely, if the system operates within an audible range, it may become annoying in hearing the stimuli. Additionally, these parameters are closely linked to authentication performance. In future work, we will address this challenge and aim to find an optimal balance among these factors.

7 Conclusion

In this paper, we propose EarID, a learning-free user authentication system utilizing ear canal scanning biometrics. EarID transforms informative cepstrum features into fixed-length binary ECS codes, enabling template matching via a fuzzy commitment scheme without requiring machine learning classifiers. Operating within a client-server architecture, EarID ensures user data is processed locally on the client device, with server guidance, addressing security and privacy concerns in remote authentication. Our evaluation results demonstrate that EarID significantly enhances efficiency while achieving a balanced accuracy of 98.1%, comparable to machine-learning classifier counterparts. During dataset scaling, EarID reduces enrollment time by 3.8× and decreases the server storage footprint per user enrollment by 22×. Additionally, communication data per authentication attempt is reduced by 31.3× compared to raw FFT features.

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References

- [1] Andy Adler, Richard Youmaran, and Sergey Loyka. 2006. Towards a measure of biometric information. In *2006 Canadian conference on electrical and computer engineering*. IEEE, New York, NY, USA, 210–213.
- [2] Amazon. 2024. *Alexa Smart Home*. Amazon. <https://www.amazon.com/alexa-smart-home/>
- [3] Frédéric Bimbot, Jean-François Bonastre, Corinne Fredouille, Guillaume Gravier, Ivan Magrin-Chagnolleau, Sylvain Meignier, Teva Merlin, Javier Ortega-García, Dijana Petrovska-Delacrétaz, and Douglas A Reynolds. 2004. A tutorial on text-independent speaker verification. *EURASIP Journal on Advances in Signal Processing* 2004 (2004), 1–22.
- [4] Raj Chandra Bose and Dwijendra K Ray-Chaudhuri. 1960. On a class of error correcting binary group codes. *Information and control* 3, 1 (1960), 68–79.
- [5] Yetong Cao, Chao Cai, Anbo Yu, Fan Li, and Jun Luo. 2023. EarACE: Empowering Versatile Acoustic Sensing via Earable Active Noise Cancellation Platform. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 7, 2 (2023), 1–23.
- [6] CUI Devices. 2024. DigiKey Page: CMC-3015-44L100 Product. <https://www.digikey.com/en/products/detail/cui-devices/CMC-3015-44L100/7398899> Accessed: 2024-05-25.
- [7] Xiaoran Fan, Longfei Shangguan, Siddharth Rupavatharam, Yanyong Zhang, Jie Xiong, Yunfei Ma, and Richard Howard. 2021. HeadFi: bringing intelligence to all headphones. In *Proceedings of the 27th Annual International Conference on Mobile Computing and Networking*. ACM New York, NY, USA, New York, NY, USA, 147–159.
- [8] Robert Gallager. 1962. Low-density parity-check codes. *IRE Transactions on information theory* 8, 1 (1962), 21–28.
- [9] Yang Gao, Wei Wang, Vir V Phoha, Wei Sun, and Zhanpeng Jin. 2019. EarEcho: Using ear canal echo for wearable authentication. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 3, 3 (2019), 1–24.
- [10] Apple Inc. 2023. *About Touch ID advanced security technology*. Apple Inc. <https://support.apple.com/en-us/105095>
- [11] Andrew Teoh Beng Jin, David Ngo Chek Ling, and Alwyn Goh. 2004. Biohashing: two factor authentication featuring fingerprint data and tokenised random number. *Pattern recognition* 37, 11 (2004), 2245–2255.
- [12] Ari Juels and Martin Wattenberg. 1999. A fuzzy commitment scheme. In *Proceedings of the 6th ACM conference on Computer and communications security*. ACM, New York, NY, USA, 28–36.
- [13] Pınar Kürtünlioğlu, Beste Akdik, and Enis Karaarslan. 2022. Security of virtual reality authentication methods in metaverse: An overview. *arXiv preprint arXiv:2209.06447* (2022), 1–5.
- [14] Shivangi Mahto, Takayuki Arakawa, and Takafumi Koshinaka. 2018. Ear acoustic biometrics using inaudible signals and its application to continuous user authentication. In *2018 26th European Signal Processing Conference (EUSIPCO)*. IEEE, New York, NY, USA, 1407–1411.
- [15] Davide Maltoni, Dario Maio, Anil K Jain, Salil Prabhakar, et al. 2009. *Handbook of fingerprint recognition*. Vol. 2. Springer, New York, NY, USA.
- [16] Kren Rahbek Nørgaard, Karolina K Charaziak, and Christopher A Shera. 2019. A comparison of ear-canal-reflectance measurement methods in an ear simulator. *The Journal of the Acoustical Society of America* 146, 2 (2019), 1350–1361.
- [17] A.V. Oppenheim, A.S. Willsky, and S.H. Nawab. 1997. *Signals & Systems*. Prentice Hall, Upper Saddle River, NJ. <https://books.google.com/books?id=LwQQAQAAIAAJ>
- [18] Nobuyuki Otsu. 1979. A Threshold Selection Method from Gray-Level Histograms. *IEEE Transactions on Systems, Man, and Cybernetics* 9, 1 (1979), 62–66. <https://doi.org/10.1109/TSMC.1979.4310076>
- [19] PayByFace. 2024. *PayByFace – Face Recognition Payment System*. PayByFace Company. <https://paybyface.io>
- [20] Sony. 2024. Amazon Page: Sony MDREX15AP/B Black Earbuds. <https://www.amazon.com/Sony-MDREX15AP-B-Black/dp/B00JG2WRUO> Accessed: 2024-05-25.
- [21] Sophie Stephenson, Bijeeta Pal, Stephen Fan, Earlene Fernandes, Yuhang Zhao, and Rahul Chatterjee. 2022. Sok: Authentication in augmented and virtual reality. In *2022 IEEE Symposium on Security and Privacy (SP)*. IEEE, New York, NY, USA, 267–284.
- [22] Susan E Voss, Gabrielle R Merchant, and Nicholas J Horton. 2012. Effects of middle-ear disorders on power reflectance measured in cadaveric ear canals. *Ear and hearing* 33, 2 (2012), 195–208.
- [23] Zi Wang, Sheng Tan, Linghan Zhang, Yili Ren, Zhi Wang, and Jie Yang. 2021. Eardynamic: An ear canal deformation based continuous user authentication using in-ear wearables. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 5, 1 (2021), 1–27.
- [24] Stephen B Wicker and Vijay K Bhargava. 1999. *Reed-Solomon codes and their applications*. John Wiley & Sons, Minneapolis, MN, USA.
- [25] Masaki Yasuhara, Isao Nambu, and Shohei Yano. 2022. Bilateral ear acoustic authentication: A biometric authentication system using both ears and a special earphone. *Applied Sciences* 12, 6 (2022), 3167.
- [26] Wenyi Zhao, Rama Chellappa, P Jonathon Phillips, and Azriel Rosenfeld. 2003. Face recognition: A literature survey. *ACM computing surveys (CSUR)* 35, 4 (2003), 399–458.
- [27] Hui Zhong, Chenpei Huang, Xinyue Zhang, and Miao Pan. 2023. Metaverse CAN: Embracing Continuous, Active, and Non-intrusive Biometric Authentication. *IEEE Network* 37, 6 (2023), 67–73.